# INSPECTION AND TEST PLAN (ITP) - 48" SAWH API 5L X65 PSL2 LINE PIPE

| Item | Test / Inspection Description | Frequency / Quantity | Acceptance Criteria / Standard |
|---|---|---|---|
| 1 | Ladle Heat Analysis | One analysis per heat of steel | C <= 0.12%, P <= 0.020%, S <= 0.010%, CEpcm <= 0.22 |
| 2 | Product Chemical Analysis | Two analyses per heat of steel | API Spec 5L PSL2 Table 5 Limits |
| 3 | Pipe Body Tensile Test | Once per test unit (lot) of 100 pipes | Rt0.5 >= 450 MPa, Rm: 535-760 MPa, Af >= 19.5% |
| 4 | Weld Seam Tensile Test | Once per test unit (lot) | Rm >= 535 MPa across weld seam |
| 5 | Charpy V-Notch (CVN) Body Impact | 1 set (3 specimens) per lot at 0 °C | Min Avg 41 Joules, Min Individual 31 Joules |
| 6 | Charpy V-Notch (CVN) Weld & HAZ | 1 set weld + 1 set HAZ per lot at 0 °C | Min Avg 27 Joules, Min Individual 20 Joules |
| 7 | Drop Weight Tear Test (DWTT) | Once per heat (2 specimens) at 0 °C | Average shear area >= 85% |
| 8 | Guided-Bend Test | 1 root + 1 face bend per test unit | No defect or opening > 3.2 mm in weld/HAZ |
| 9 | Mill Hydrostatic Pressure Test | Each pipe (100%) | Test pressure 110 bar, 5 seconds holding time |
| 10 | Ultrasonic Weld Seam Testing (UT) | 100% full length of weld seam | API 5L Annex E / ISO 10893-11 Level U2 |
| 11 | Pipe Ends UT Laminar Testing | 100% of pipe ends (100 mm band) | API 5L Annex E.8 (No defect > 6 mm) |
| 12 | Diameter and Out-of-Roundness | Once per 4 hours and 100% pipe ends | Body dia +/- 0.5%, End ovality <= 1.0% D |
| 13 | Wall Thickness Verification | 100% of all pipes (Ultrasonic / Gauge) | Tolerance: -8.0% / +15.0% (API 5L Table 11) |
| 14 | Visual Surface Inspection | 100% internal and external surface | No crack, laminate, sliver or defect > 12.5% t |
| 15 | Residual Magnetism Measurement | At least once per 4 hours on pipe ends | Average <= 3.0 mT (30 Gauss), Max <= 3.5 mT |